



Torch Height Control Manual

Read This Before You Begin... (The proverbial “fine print”)

*Any computer controlled machine tool is potentially dangerous. You’ve already accepted responsibility for the operation of the CNC machine you expect to control with the THC. The THC is a device that was designed to control the Z axis height while running the Mach © software and cutting metal using a plasma cutter.

The Mach software runs a myriad of different types of machines worldwide. This manual is here to give you guidance on safety issues and proper operation of the THC and associated options. Since we do not know the details of your machine, environment, or operator skill level, we cannot take responsibility for any damage or injury caused by its use. It is your responsibility that you understand your machine, its capabilities, and any safety issues it generates.

If you are in any doubt, be sure to seek guidance from a professionally qualified expert rather than risk injury to yourself, others, or damage to your equipment.

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Definitions

*Definitions and terminology are important as it gives the developers and users a basis in their understanding THC function and operation. Below are several terms we use in this document and the definitions as we see them.

Torch Height Control

*Plasma machines use a constant current power supply for cutting. The power supply can range from 10 to 200 amps (or more) selected from the plasma machine control panel. A constant current supply keeps the cutting current at the user selected value by changing the cutting voltage as the load changes. The load changes as the distance from the torch tip to the metal you are cutting changes. The THC301d torch height control monitors this “Tip Voltage” and adjusts the torch height to maintain Tip Voltage to within + or – 0.5v of the user Set Voltage.

Cut Height

*The Cut Height is the optimum distance from the tip of the torch to the top of the metal being cut while cutting.

Pierce Height

*The Pierce Height is approximately 1.5 times the distance of the Cut Height. The torch is set to the Pierce Height when the torch is about to fire and pierce the metal to begin cutting. This extra distance helps prevent molten metal blowback from damaging the torch tip when the torch is fired and you are piercing the metal.



Tip Volts or Arc Volts

*Tip Voltage or Arc Voltage is the voltage measured from the tip of the torch to the metal while being cut. This voltage can range from 0 to 255 volts DC. In special cases, the THC301d can be calibrated to reach up to 512vdc. This voltage is directly proportional to the distance between the torch tip and the metal being cut.

Set Voltage or Target Voltage

*The Set Voltage is the “target” voltage you wish to have the THC maintain while cutting and can be adjusted from the front panel of the THC301d. The Set Voltage (adjusted by the user using the knob on the front panel of the THC) is compared to the Tip voltage and the THC tells Mach to adjust the Z axis up or down until the Tip voltage equals the Set voltage. Maintaining this Tip Voltage insures a constant distance between the tip and the metal being cut. This is the main purpose of the THC.

Resolution

*The THC301 will only display readings in full volts (not fractions), but the internal resolution is kept in fractions.

Control Program

*The THC301d was designed to work with the Mach3 control program. The “X, Y, Z, and A” axis are controlled from parallel port 1 and the THC301d works from parallel port 2.

Hall Effect Current Sensor – HECS

*Most current models of plasma cutters will provide a signal that tells the control program (or THC) when the arc is established and ready to start cutting. This signal called “Arc OK”, “Arc Good”, or “OK-to-Move” will be connected to the Sensor Board and it will relay the information to the THC. In some older models of plasma cutters, this signal is not readily available. The HECS uses a Hall Effect sensor embedded in a ferrite core and clamped around the “work” or ground cable to detect when the plasma arc has been established and the torch is ready to start cutting. This HECS generates the “Arc OK” signal on the Sensor Board

General Description

The simplest explanation for the purpose of a Torch Height Controller is this.

1. A plasma cutter’s torch cuts metal while being positioned a precise distance above the metal it is cutting.
2. This **optimum** distance is determined by many factors. These factors include:
 - Type of metal being cut (mild steel, stainless steel, high carbon steel, and even aluminum)
 - Thickness of metal. Anywhere from thin sheet to material several inches thick.
 - Rate-of-cut measured in inches or millimeters per second or minute.
 - Cutting amperage (as low as 10 to 20 amps up to 200 or more)
3. This “optimum” distance is desired for many reasons including these:
 - Least amount of dross or slag on the bottom of the cut.
 - Finest kerf (or slot diameter). Think of this as equivalent to “blade width” if cutting with a saw.



- Least amount of beveling while cutting turns (corners). Beveling is other than a cut 90 degrees (perpendicular) to the material.
- Consumable (Tip and electrode) longevity!

THC301d System Requirements and Preferences

Mach3 - Required

The THC301d **is not a standalone** Torch Height Controller.

At the present, it requires **a licensed copy** of Mach3 as the control program.

Second Parallel Port - Required

The THC301d is connected to a second parallel port on the PC. If your PC does not have a second parallel (printer) port, they can be purchased from many places (including eBay) for less than \$20.00. Mach3 supports the second parallel port and uses it to control the THC301d.

SheetCam – Highly Recommended

Though not really required to run the THC, SheetCam will generate the G-code necessary to turn the torch on/off whereas many G-code generators will only provide the G-code for a router which turns on at the beginning of the G-code and turns off at the end. SheetCam provides much more than this including touch-off, and torch specific operations.

Floating Head Z Axis Switch for Touch-Off – Preferred

At the present, most THC301d users have what is called a “floating head” on their Z axis that holds the torch. This Floating Head mechanism provides slide movement of .25 inch or less between the Z axis and the Torch. A switch is connected such that it activates at the top end of the floating head movement. This switch is configured in Mach and is used to automatically “touch-off” the torch and set it at the preferred height for piercing and cutting the metal. Though not required, as the torch can be manually set above the metal for piercing and when the arc is established, the THC will take over operations, it is very much a time saver and automates the piercing especially when the metal is not perfectly flat and piercing distance in different areas is not consistent.





Misconceptions of Requirements for use of the THC301d

Use of the THC301d is **not** dependent on:

- Use of any special electronics or motors to drive your gantry as long as they can be controlled by the Mach3 control program.
- An "Arc OK" or "Arc Good" signal from your plasma cutter. If your plasma cutter does not provide this signal (many "hand torch only" systems do not), we provide a Hall Effect Current Sensor (HECS) module that clamps around the ground or "work cable" and then plugs into our Sensor Board to provide this signal.



Connecting Cables

There are 3 cables used in the THC301.

- The Sensor cable used to connect the THC Logic Board to the Sensor Board
- PC Port 2 cable used to connect the THC to the PC (and Mach3)
- The HECS cable (if needed) to connect the HECS to the Sensor Board

Sensor Cable

*The Sensor Cable is a DB9 male to female 25ft long cable that connects from the THC Logic Board to the Sensor Board. It provides communication between the THC and the plasma machine. It cannot be connected wrong, but make sure that you screw in the holding screws to provide a solid connection that can't be vibrated loose.

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9 Pin Cable



PC Port 2 Cable

*The PC parallel port cable is a DB25 (full pin) male to male cable that connects between the PC second parallel (printer) port card and the THC. Make sure the cable you use for this connection is a “full pin” straight through cable which means that all 25 pins have corresponding wires in the cable and are connected “one for one” (i.e. pin 1 of one end connects pin 1 of the other, etc.) Make sure your cable is designed to be used as a printer cable



25 Pin Cable (Port 2)



Mach Screen THC Function Definitions

Torch Height Control Area (Center Bottom)

***Torch On/Off** – This is both a “pushbutton” and indicator. It allows manually firing the torch by clicking on the Torch button and the red bar above the button will light. When Mach3 send the Torch On signal to the THC, the red bar also lights.

***THC On/Off** – This is the control activating the THC. When it is off, Mach3 ignores any commands from the THC. When it is on, Mach3 will depend on the THC for communication to maintain proper tip height of the torch.

***Up – Down** – These indicators show when the Z axis is moving Up or Down.

***ARC OK** – This indicator shows when the THC reports to Mach3 that the Arc has transferred and Arc OK is valid.

***Torch Active** – I have no idea what this indicator does other than, possibly, the obvious.

Feed Rate Area (Right of Torch Height Control)

***THC Rate** – This number is a **percentage** of the set cutting feed rate and is used to throttle the Z axis movement and make it proportional to the feed rate. If the THC Rate is set to low, the Z axis will not keep up with sudden Arc Voltage changes such as when cutting corrugated material. If the THC Rate is set too high the Z axis becomes overly sensitive to Arc Voltage changes and the Z axis will oscillate up and down trying to the axis perfectly “in the zone”.

Settings Area (Right Bottom)

***Max THC** – This is the maximum limit the Z axis will be allowed to travel under THC control

***Min THC** – This is the minimum limit the Z axis will be allowed to travel under THC control.

Torch Height Control Operation Sequence for THC301d

Assuming Mach3 is set up properly for THC Plasma operation and a G-Code “cut file” has been properly generated from SheetCam, the sequence of events follows when plasma cutting:

- 1.** The Plasma Torch is at some “retract” height above the work metal.
- 2.** Pressing Start on Mach3 (with the THC On/Off button activated on the Mach3 screen under Torch Height Control) causes the torch to move to the requested X/Y position. The Z axis moves down and the “floating head switch” or ohmic sensor detects the work piece with the tip of the torch.



- 3.** Mach3 then raises the torch to the Pierce Height (determined by settings in SheetCam) – P=Pierce Height
- 4.** Mach3 then turns on the TORCH light under Torch Height Control and tells the THC to fire the torch (Torch On signal). The THC turns on the TORCH ON LED on the front panel then sends the signal to the THC Sensor board where the relay activates and sends the “torch on” signal to the plasma cutter. Gas flow (shop air in most cases) begins.
- 5.** The plasma pilot arc initiates via the plasma machine. The pilot arc transfers to the cutting arc and cutting arc connects with the work piece. Arc OK signal is sent to the THC Sensor board via either an on-board “Arc OK or “Arc Good” signal from the plasma cutter or the Hall Effect Current Sensor detects a current spike (when the transfer from pilot arc to cutting arc is executed) and the Green LED on the THC Sensor board lights. The “Arc OK” signal is then transferred to the THC where it turns on the ARC OK LED on the THC front panel and then sends it to Mach3 via the parallel port cable. Mach3 then turns on the ARC OK square “LED” on the screen in the Torch Height Control section. If there is a Pierce Time Delay that was set in SheetCam for this operation, then the Mach3 delays moving the torch until the delay has ended.
- 6.** Now Mach3 working with the THC moves the torch down to the Cut Height as determined by the Arc Voltage. The Arc Voltage or Tip Voltage is sensed by the Voltage Divider Circuit on the THC Sensor board and sent to the THC. The THC compares it to the Set Voltage (Set Voltage is the number the user set in the THC by adjusting the TIP HEIGHT knob(s) on the THC front panel and reflected in the THC display) and tells Mach3 to adjust the Z axis to compensate. Mach3 now tells the torch to move and start cutting.
- 7.** As the torch moves and cuts the work piece the Tip Voltage may change if the metal warps or is not perfectly flat. This necessitates movement of the Z axis to maintain the proper cut height. The THC tells Mach3 to move the Z axis up if the Set Voltage is higher than the Tip Voltage or down if the Set Voltage is higher than the Tip Voltage. If they are equal, then no Z axis motion is performed. These actions are reflected in both the UP and DOWN lights on the Mach3 screen and on the THC.
- 8.** When the cut is finished, Mach3 stops X/Y movement, tells the THC to turn the torch off, whereas the THC loses Arc OK, and Mach3 raises the Z axis to the “retract” height. If there is more to cut, then Mach3 will move the torch to the next X/Y position and the process will start over.